MILITARY SPECIFICATION SHEET

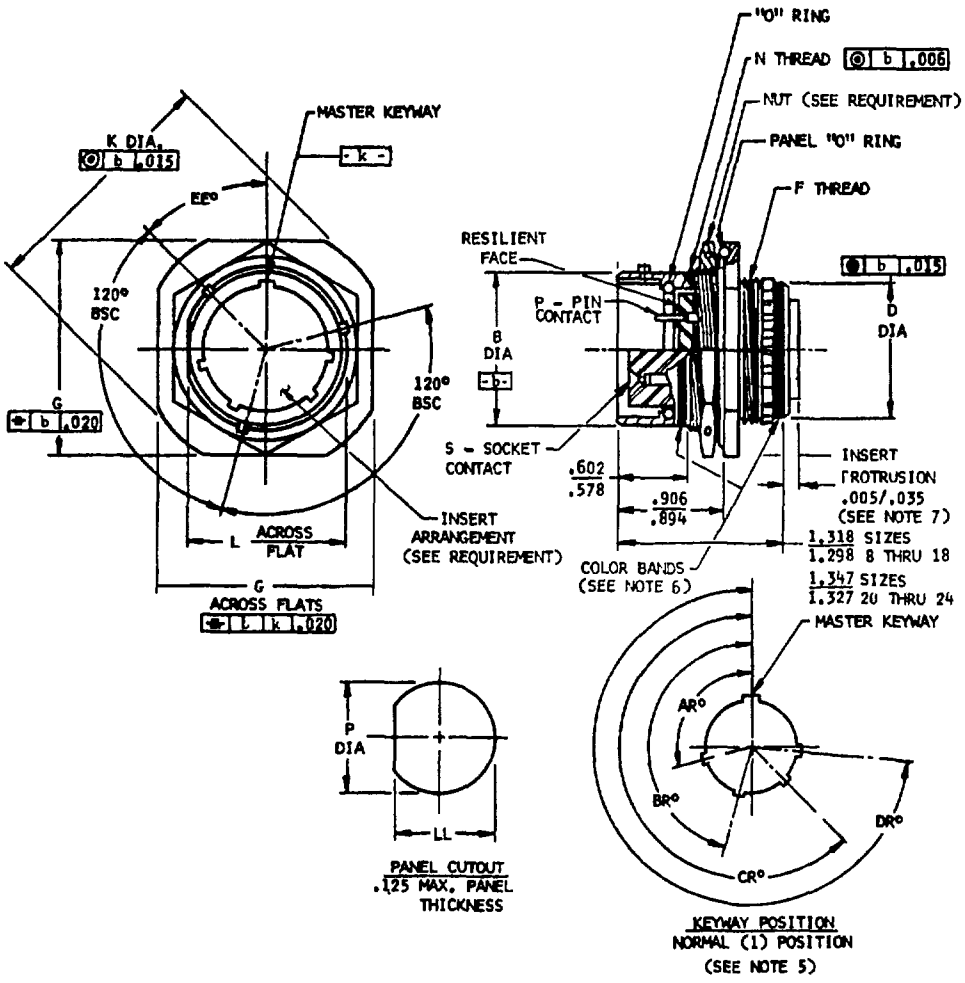
CONNECTORS, ELECTRICAL, CIRCULAR, HIGH DENSITY, QUICK DISCONNECT, ENVIRONMENT RESISTING: RECEPTACLE, JAM NUT MOUNT, INDIVIDUAL RELEASE, CRIMP TYPE CONTACTS CLASS A, F AND W (SERIES 3)

Ⓐ

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the connectors described herein shall consist of this document and the latest issue of Specification MIL-C-81511.

THIS SPECIFICATION INACTIVE FOR NEW DESIGN AFTER 30 APRIL 1975. SEE MIL-STD-1353.



Ⓐ denotes changes

FSC 5935

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TABLE I.

SHELL SIZE	B DIA	D DIA	F THREAD	G ACROSS FLATS	K DIA	L ACROSS FLATS	N THREAD	P DIA	LL
8	.553 (14.05) .547 (13.89)	.430 (10.92) .424 (10.77)	1/2-28 UNEF-2A	.989 (25.12) .969 (24.61)	1.072 (27.23) 1.052 (26.72)	.598 (15.19) .590 (14.99)	5/8-20 UN-2A	.640 (16.26) .630 (16.00)	.610 (15.49) .600 (15.24)
10	.678 (17.22) .672 (17.07)	.555 (14.10) .549 (13.94)	5/8-28 UN-2A	1.114 (28.30) 1.094 (27.79)	1.197 (30.40) 1.177 (29.90)	.723 (18.36) .715 (18.16)	3/4-20 UNEF-2A	.765 (19.43) .755 (19.18)	.735 (18.67) .725 (18.42)
14	.928 (23.57) .922 (23.42)	.805 (20.45) .799 (20.29)	7/8-28 UN-2A	1.401 (35.59) 1.381 (35.08)	1.510 (38.35) 1.490 (37.85)	.973 (24.71) .965 (24.51)	1-20 UNEF-2A	1.015 (25.78) 1.005 (25.53)	.985 (25.02) .975 (24.77)
16	1.054 (26.77) 1.048 (26.62)	.930 (23.62) .924 (23.47)	1-28 UN-2A	1.526 (38.76) 1.506 (38.25)	1.635 (41.53) 1.615 (41.02)	1.098 (27.89) 1.090 (27.69)	1 1/8-20 UN-2A	1.140 (28.96) 1.130 (28.70)	1.110 (28.19) 1.100 (27.94)
18	1.178 (29.92) 1.172 (29.77)	1.055 (26.80) 1.049 (26.64)	1 1/8-28 UN-2A	1.651 (41.94) 1.631 (41.43)	1.745 (44.32) 1.725 (43.82)	1.222 (31.04) 1.214 (30.84)	1 1/4-20 UN-2A	1.265 (32.13) 1.255 (31.88)	1.235 (31.37) 1.225 (31.12)
20	1.303 (33.10) 1.297 (32.94)	1.180 (29.97) 1.174 (29.82)	1 1/4-28 UN-2A	1.776 (45.11) 1.756 (44.60)	1.870 (47.50) 1.850 (46.99)	1.347 (34.21) 1.339 (34.01)	1 3/8-20 UN-2A	1.390 (35.31) 1.380 (35.05)	1.360 (34.54) 1.350 (34.29)
22	1.428 (36.27) 1.422 (36.12)	1.305 (33.15) 1.299 (32.99)	1 3/8-28 UN-2A	1.901 (48.29) 1.881 (47.78)	1.995 (50.67) 1.975 (50.17)	1.472 (37.39) 1.464 (37.19)	1 1/2-20 UN-2A	1.515 (38.48) 1.505 (38.23)	1.485 (37.72) 1.475 (37.47)
24	1.553 (39.45) 1.547 (39.29)	1.430 (36.32) 1.424 (36.17)	1 1/2-28 UN-2A	2.026 (51.46) 2.006 (50.95)	2.120 (53.85) 2.100 (53.34)	1.597 (40.56) 1.589 (40.36)	1 5/8-20 UN-2A	1.640 (41.66) 1.630 (41.40)	1.610 (40.89) 1.600 (40.64)

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TABLE I. - CONTINUED

SHELL SIZE	AR° BSC	BR° BSC	CR° BSC	DR° BSC	PANEL "O"RING	EE° BSC - POSITION					
						1	2	3	4	5	6
8	105	140	215	265	MS9068-018	60	46	43	60	89	43
10	95	141	208	236	MS9068-020	53	87	51	77	87	87
14	95	141	208	236	MS9068-024	53	87	51	77	87	87
16	80	142	196	293	MS9068-026	98	98	98	98	98	98
18	80	142	196	293	MS9068-028	98	98	98	98	98	98
20	80	142	196	293	MS9068-029	98	98	98	98	98	98
22	80	142	196	293	MS9068-030	98	98	98	98	98	98
24	80	142	196	293	MS9068-031	98	98	98	98	98	98

NOTES.

1. DIMENSIONS ARE IN INCHES.
2. MILLIMETERS ARE IN PARENTHESES.
3. METRIC EQUIVALENTS (TO THE NEAREST .01 MM) ARE GIVE FOR GENERAL INFORMATION ONLY AND ARE BASED UPON 1 INCH = 25.4 MM.
4. DIMENSIONS ARE AFTER PLATING.
5. NORMAL MATING KEYWAY POSITION SHOWN. FOR ALTERNATE KEYWAY POSITIONS SEE MIL-C-81511, FIGURE 8.
6. COLOR BANDS: COLOR BLUE .100 MAX. WIDE, MUST BE VISIBLE WHEN MOUNTED.
7. ARRANGEMENTS CONTAINING SIZE 12 CONTACTS, INSERT PROTRUSION .070/.020.

INCHES	MM	INCHES	MM
.003	.08	.230	5.84
.005	.13	.578	14.68
.006	.15	.602	15.29
.015	.38	.894	22.71
.020	.51	.906	23.01
.067	1.70	1.298	32.97
.073	1.85	1.318	33.48
.090	2.29	1.347	33.71
.094	2.39	1.327	34.21
.100	2.54	.035	.89
.125	3.18		

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REQUIREMENTS:

Dimensions and configuration: See figure and table I.

Series: 3

Ⓐ Classes. A, F and W.

Shell size. See table II.

Insert arrangement: See supplement for MS sheet corresponding to desired shell size.

Contacts: Style P or S, crimp type.

Shell polarization: 1 thru 6, key position.

Weight: See table II.

Marking: Part number on shell.

Mating counterpart: Plug connectors specified in MIL-C-81511/26, /38, and /46.

Insertion and removal tool: As specified in MS3160.

Slotted nut: Shall conform to MIL-C-81511/20.

Part number: The part number shall consist of M81511/49 together with the applicable codes for class, shell size, insert arrangement, contact style, and key positions. Information concerning the formulation of the part number can be found in MIL-C-81511.

TABLE II. Shell size and weight.

Shell size	Weight (ref) 1/ (pounds)
24 (Code J) 22 (Code H) 20 (Code G) 18 (Code F) 16 (Code E) 14 (Code D) 10 (Code B) 8 (Code A)	To be determined

1/ Removable crimp contacts not included in weight. (See contact drawings for weight of pin and socket contact.)

Custodian.
Navy - AS

Review activities.
Navy - EC, OS, SH
DSA - ES

User activities:
Navy - MC

Preparing activity:
Navy - AS

Agent:
DSA - ES.

(Project 5935-N023-6)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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INSTRUCTIONS. The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-C-81511/49A(Navy) Connectors, Electrical, Circular, High Density, Quick Disconnect

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A. GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?
☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

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